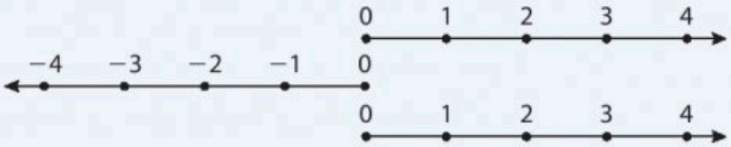
Investigation:

Considering integers as directed numbers, investigate the principle involved in each of the following lines, particularly line (iii)

(i) $+4$ ...
(ii) -4 ...
(iii) $-(-4)$ ...



Try to find a practical example to illustrate (iii)

Exercise 2.3

1. Write down the answer to each of the following:

(i) 6×4 (ii) -5×4 (iii) $7 \times (-8)$
(iv) $5 \times (-9)$ (v) -6×9 (vi) $(-9) \times (-4)$
(vii) $(-8) \times (-6)$ (viii) $10 \times (-7)$ (ix) -9×8
(x) $(-6) \times (-11)$ (xi) $4 \times (-20)$ (xii) $(-18) \times 3$

2. Work out each of these:

(i) $-12 \div 2$ (ii) $36 \div (-4)$ (iii) $-20 \div 5$
(iv) $-15 \div (-5)$ (v) $-30 \div (-10)$ (vi) $-24 \div 8$

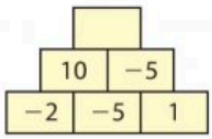
3. Write down the answer to each of these:

(i) $\frac{16}{2}$ (ii) $\frac{-8}{4}$ (iii) $\frac{-27}{3}$ (iv) $\frac{15}{-3}$ (v) $\frac{-42}{-7}$
(vi) $\frac{-35}{7}$ (vii) $\frac{54}{-6}$ (viii) $\frac{-54}{-9}$ (ix) $\frac{-63}{7}$ (x) $\frac{55}{-11}$

4. Find the missing number in each of the following:

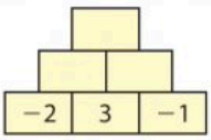
(i) $6 \times \square = -48$ (ii) $-4 \times \square = 24$ (iii) $-5 \times \square = -40$
(iv) $-20 \div \square = -4$ (v) $36 \div \square = -9$ (vi) $\square \div (-7) = -3$

5. This is a 'multiplication wall'.
The number on each brick is found by multiplying the two numbers on the bricks below.
What will be the number on the top brick of this wall?

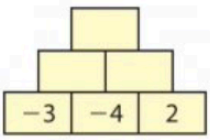


6. Copy and complete these multiplication walls.

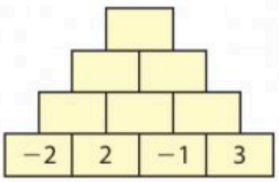
(i)



(ii)



(iii)



7. Express as a single integer:

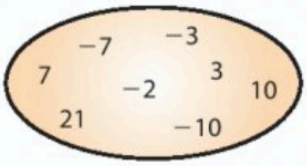
(i) $6 \times 2 \times (-3)$ (ii) $4 \times (-3) \times (-2)$ (iii) $(-6) \times 5 \times (-3)$
(iv) $(-4) \times (-3) \times 5$ (v) $(-2) \times 6 \times (-8)$ (vi) $(-3) \times (-2) \times (-7)$

8. Express each of the following as a single integer:

(i) $\dfrac{3 \times (-6)}{9}$ (ii) $\dfrac{(-5) \times (-10)}{-10}$ (iii) $\dfrac{(-12) \times 4}{8}$
(iv) $\dfrac{(-8) \times (-9)}{-12}$ (v) $\dfrac{9 \times (-6)}{-2}$ (vi) $\dfrac{(-8) \times 3 \times (-6)}{-12}$

9. Choose pairs of numbers from the loop to make these calculations correct.

(i) $\square \div \square = -3$ (ii) $\square \times \square = 14$
(iii) $\square \times \square = -6$ (iv) $\square \div \square = -5$
(v) $\square \div \square = 5$ (vi) $\square \times \square = -63$



10. Given that $(-4)^2 = (-4) \times (-4) = 16$, work out each of these:

(i) $(-3)^2$ (ii) $(-6)^2$ (iii) $(-7)^2$ (iv) $(-2)^3$

11. Noting that you multiply and divide before you add and subtract, simplify each of the following:

(i) $3 - 8 + 8 \times (-2)$ (ii) $15 + 6 \div 2 + 12 \times (-2)$
(iii) $(16 - 4) \div 2 + 5 \times (-4)$ (iv) $(15 - 10) \times -3 + 5 \times (-6)$
(v) $14 + 4 \times (-2) - (7 - 15)$ (vi) $100 \div (13 - 3) + (-4) \times (-7)$.

12. Since $3 \times 4 = 12$, we call 12 the product of the two numbers 3 and 4.
Now copy and complete the table shown below.

	Sum	Product	Two numbers
(i)	−7	10	
(ii)	−13	30	
(iii)	−5	6	
(iv)	5	−6	
(v)	−8	12	
(vi)	−2	−15	